

LAB MODULE #3 - PULSTAR REACTOR STARTUP

Objective: Predict and observe reactor behavior while performing a routine reactor startup. Take the reactor from a shutdown state through sub-critical multiplication and to super-criticality. Establish critical rod positions.

Characteristics of **SUB-CRITICAL MULTIPLICATION**:

- ***Doubling the Source Range Monitor (SRM) count-rate*** halves the difference between the ***current value for K_{eff} and 1***, therefore ***taking the reactor halfway to criticality***.
- ***Withdraw the control rods and double the SRM count-rate approximately 5 times*** to bring the reactor to criticality (or just super-critical).
- ***The closer you get to criticality, the longer it takes*** for the SRM count rate to ***reach subcritical equilibrium*** (i.e. a stable non-increasing neutron population).

Characteristics of a **SUPER-CRITICAL REACTOR**:

- ***Linearly increasing traces on the SRM and Log-N semi-log plot recorder traces*** (without control rod motion) indicates an exponential increase in the reactor neutron population. This is an indication that the reactor is ***SUPER-CRITICAL***.
- A ***Positive Stable Period (τ)*** (without control rod motion) as indicated on the SRM and Log-N Channel meters confirms an exponential increase in neutron population. This is another indication that the reactor is ***SUPER-CRITICAL***.

Institute for Nuclear Power Operations (INPO) - Guidelines for Reactor Operations:

- a) **STOP & THINK** – *what operations evolution/procedure is being performed?*
- b) **TOUCH** – *which control does the procedural step indicate should be manipulated?*
- c) **VERIFY** – *am I about to manipulate the correct control?*
- d) **ANTICIPATE** – *how should the reactor systems respond to the manipulation?*
- e) **MANIPULATE** – *perform the control manipulation.*
- f) **OBSERVE** – *observe the change in reactor behavior in response to the control manipulation.*

***Finally, (RE-)VERIFY that the reactor behavior you OBSERVED is what you ANTICIPATED!
Otherwise – SCRAM!!***

NOTE: *The responses of the reactor systems to correct manipulations of the controls are **FULLY UNDERSTOOD** and are **ENTIRELY PREDICTABLE!***

Useful Definitions:

- **CRITICALITY** – Neutron population is stable and maintained by fission; where $K_{eff} = 1$; $\rho = 0$
- **SUB-CRITICALITY** – neutron population is decreasing; where $0 < K_{eff} < 1$; $\rho < 0$
- **SUPER-CRITICALITY** – neutron population is increasing exponentially due to an increasing fission rate; where $K_{eff} > 1$; $\rho > 0$
- **REACTOR PERIOD (τ)(SEC)** – The time in seconds it takes for reactor power to change by a factor of 'e' (i.e. 2.718). A small value indicates power is changing quickly; a large or infinite value indicates power is changing slowly over time or not at all.
- **REACTIVITY (ρ)** – A measure of the departure from criticality; where $\rho = (K_{eff} - 1) / K_{eff}$